

CareerGuidance: A Web-Based Platform for Personalized Career Counseling and Skill Development

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Abstract—Selecting an appropriate career path remains a significant challenge for many students in India, particularly in semi-urban and rural regions, where structured guidance and mentorship are limited. This paper presents *CareerGuidance*, a web-based platform that combines a Gemini-powered AI chatbot, personalized roadmap generator, mentor-connect module, and gamified progress tracking to provide scalable, contextual, and real-time career counseling to students. The platform integrates a React + Tailwind frontend with a Django REST backend, MySQL storage, and WebSocket-based, real-time communication. We describe the system architecture, hybrid AI pipeline for intent detection and recommendation, evaluation methodology, and example synthetic results from pilot usability testing. The results indicate improved clarity in career decision-making, a high usability score in preliminary surveys, and acceptable system performance under typical classroom loads. We conclude with a discussion of the limitations, relevance to the NEP 2020 objectives, and opportunities for future work.

Index Terms—career counseling, AI chatbot, personalized learning, Gemini, WebSockets, Django, React, educational technology

I. INTRODUCTION

Career decision-making is a complex, multi-factor problem influenced by a student's aptitude, interests, socio-economic context, and the rapidly changing demands of the job market. Traditional career counseling methods are often scarce, centralized, or outdated, especially in many Indian educational contexts where access to trained counselors is limited. The National Education Policy (NEP) 2020 highlights the need for accessible, equitable, and technology-enabled career guidance [1].

CareerGuidance aims to address this gap by providing a low-cost, scalable platform that offers personalized career roadmaps, real-time interaction with mentors, and an AI-driven chatbot (Gemini-based) for 24/7 guidance. The platform is designed to be device-agnostic, multilingual, and extensible to new modules such as scholarship tracking, internships, and regional language support.

II. RELATED WORK AND LITERATURE REVIEW

AI and chatbot solutions for education and career guidance have attracted substantial research interest in recent years.

Systematic reviews have shown that chatbots improve accessibility, enable personalized interactions, and support learners in routine queries and guidance [3], [4]. Specific implementations of career guidance chatbots have been explored in several application-focused studies and prototypes [5]–[7].

Transformer-based language models, such as BERT, underpin many modern NLP pipelines for intent detection, entity extraction, and semantic matching [2]. Reviews of transformer applications emphasize both their power and the associated computational costs [8]. For deployed systems, hybrid architectures that combine rule-based components with LLMs are common; rules provide safety, deterministic routing, and low-latency responses for common scenarios, whereas LLMs handle free-text, ambiguous, or exploratory queries [3], [4].

A. Key findings from literature

- Chatbots in education increase engagement and provide round-the-clock assistance, but design quality and evaluation are crucial for meaningful outcomes [3].
- Transformer-based encoders (e.g., BERT) significantly improve intent and entity recognition tasks when fine-tuned on domain-specific data [2].
- Career guidance systems benefit from integrating labor-market signals (job-posting analytics) to suggest relevant skills and pathways; however, data freshness and bias must be managed [5].

III. PROBLEM STATEMENT

Despite the many available career tools, students frequently receive non-actionable or overly general advice. Gaps exist in:

- Personalized, stepwise roadmaps aligned with student profiles.
- Scalable access to trained mentors.
- Real-time, conversational guidance that addresses follow-up queries.
- Fresh, market-relevant suggestions that adapt to evolving job trends.

IV. PROPOSED SYSTEM

A. Design goals

The platform was designed to achieve the following goals:

- 1) **Accessibility:** Accessible across devices; straightforward UI/UX.
- 2) **Personalization:** Roadmaps generated from user profile, skills, and preferences.
- 3) **Real-time interaction:** Mentor connections and chat updates via WebSockets.
- 4) **Safety and determinism:** Rule-based modules for critical workflows, LLM for exploration.
- 5) **Extensibility:** Easy to add modules (scholarships, internships).

B. System Overview

The platform uses a client-server architecture (React frontend, Django REST backend) with MySQL as the relational store and WebSockets (Django Channels) for real-time interactions. The AI layer is implemented as a hybrid pipeline: a lightweight intent classifier and NER (BERT-based fine-tuning if data are available) combined with calls to a Gemini model for richer conversational replies and recommendation reasoning. Rule-based routing handles deterministic tasks, such as scheduling mentor sessions and serving FAQ answers.

C. Layered Architecture Model

The system follows a structured five-layer architecture to ensure modularity, scalability, and maintainability.

- 1) **Presentation Layer:** Implemented using React and Tailwind CSS, this layer manages the user interface, handles user interactions, and communicates with backend APIs via REST and WebSockets.
- 2) **Application Layer:** Built using Django REST Framework, this layer exposes API endpoints for authentication, profile management, roadmap generation, and mentor scheduling.
- 3) **Business Logic Layer:** This layer contains the recommendation engine, scoring mechanisms, roadmap generation logic, and mentor matching algorithms.
- 4) **AI Processing Layer:** Integrates the Gemini API along with an intent detection and entity recognition module to generate contextual and personalized responses.
- 5) **Data Layer:** MySQL is used as the relational database to store user profiles, mentor records, session logs, and generated roadmaps. Data access is managed through Django ORM.

D. AI Chatbot Pipeline (Gemini + Rules)

The chatbot uses a hybrid flow.

- 1) **Preprocessing:** Input text normalized, tokenized; optional speech-to-text in frontend for voice.
- 2) **Intent Detection / NER:** Lightweight model (BERT-base fine-tuned on small domain data or a rule-based

intent matcher) classifies user intent and extracts entities (skills, interests, timelines).

- 3) **Routing:** If intent is a common FAQ or scheduling, route to rule module. If exploratory or long-form, query Gemini for suggestions.
- 4) **Recommendation Engine:** Generates a personalized roadmap by mapping user profile & extracted entities to a skill-tree and learning resources (curated), returning milestones.
- 5) **Safety filter:** Rule-based checks ensure responses do not produce harmful or misleading guidance.

E. Hybrid Career Guidance Workflow Algorithm

Algorithm 1 Hybrid AI-Based Career Guidance Workflow

Require: User Query Q , User Profile P

Ensure: Personalized Career Guidance Response R

- 1: Receive user input query Q
 - 2: Preprocess query (tokenization, normalization)
 - 3: Extract Intent I and Entities E using Intent/NER module
 - 4: Retrieve User Profile P from database
 - 5: **if** I belongs to FAQ or Scheduling **then**
 - 6: Route request to Rule-Based Engine
 - 7: Generate deterministic response
 - 8: **else**
 - 9: Forward query to Gemini AI Model
 - 10: Generate contextual career suggestions
 - 11: **end if**
 - 12: Map extracted entities E with career skill repository
 - 13: Generate candidate career paths list C
 - 14: **for** each career c in C **do**
 - 15: Compute relevance score based on:
 - 16: User interests
 - 17: Current skill level
 - 18: Career demand trends
 - 19: **end for**
 - 20: Rank career paths based on relevance score
 - 21: Generate personalized roadmap with milestones
 - 22: Apply safety and validation filters
 - 23: Return final response R to user interface
-

F. Personalized Roadmap Generation Algorithm

Algorithm 2 Career Roadmap Generation

Require: Selected Career Path C , User Skills S_u **Ensure:** Learning Roadmap L

- 1: Retrieve required skill set S_c for career C
 - 2: Identify Skill Gap $G = S_c - S_u$
 - 3: Divide G into learning phases:
 - 4: Beginner Level
 - 5: Intermediate Level
 - 6: Advanced Level
 - 7: **for** each skill in G **do**
 - 8: Map skill to curated learning resources
 - 9: Assign estimated completion timeline
 - 10: **end for**
 - 11: Organize resources into milestone structure
 - 12: Return roadmap L
-

G. Mentor Matching Algorithm

Algorithm 3 Mentor Recommendation Process

Require: User Career Interest C , Availability T **Ensure:** Recommended Mentor List M

- 1: Retrieve mentor database
 - 2: Filter mentors having expertise in career C
 - 3: Filter mentors available in time slot T
 - 4: **for** each mentor **do**
 - 5: Calculate mentor compatibility score based on:
 - 6: Experience level
 - 7: User ratings
 - 8: Domain relevance
 - 9: **end for**
 - 10: Rank mentors based on compatibility score
 - 11: Return mentor list M
-

The time complexity of the recommendation ranking step depends on the number of candidate careers available. The ranking process requires sorting operations, resulting in an approximate complexity of $O(n \log n)$.

V. IMPLEMENTATION DETAILS

A. Frontend

It was built using React and Tailwind CSS. The key pages included Home, Profile, Chatbot, Roadmap, Mentor Connect, and Admin Dashboard. The chatbot UI supports text and voice inputs (browser Web Speech API) and displays suggested resources and milestones.

B. Backend

Django REST provides endpoints for authentication, profiles, roadmap generation, mentor scheduling and analytics. Django Channels power WebSocket endpoints for real-time chat and mentor notifications. MySQL stores normalized tables: users, profiles, roadmaps, sessions, mentor_slots.

C. AI Layer

The intent/NER module can be a fine-tuned BERT (if data are available) or an off-the-shelf lightweight classifier. Gemini is used via API for conversational replies and for ranking candidate career paths. All LLM responses were post-processed using rule-based filters and combined with curated resource suggestions.

VI. SECURITY AND PRIVACY CONSIDERATIONS

Since the platform handles sensitive student data, security was a primary design concern.

A. Authentication and Authorization

JWT-based authentication ensures stateless session handling. Role-based access control differentiates students, mentors, and administrators.

B. Data Protection

Sensitive profile data are encrypted at rest using AES-based encryption. API endpoints are protected via HTTPS and CORS policies.

C. LLM Safety Guardrails

All LLM responses are passed through:

- Keyword filtering
- Prompt validation rules
- Response length limitation

VII. SYSTEM DIAGRAMS

Here, we include the ER Diagram, Flowchart, and Class Diagram as PNG images to illustrate the platform's architectural and logical structure.

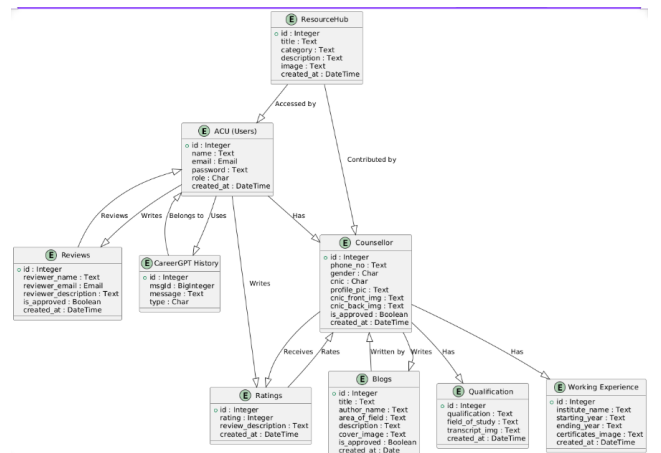


Fig. 1: Entity-Relationship Diagram of CareerGuidance platform (PNG)

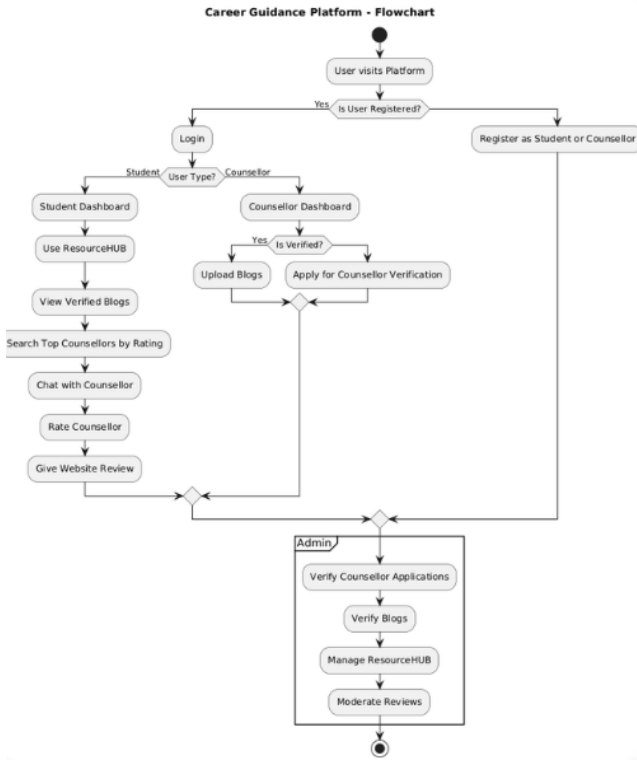
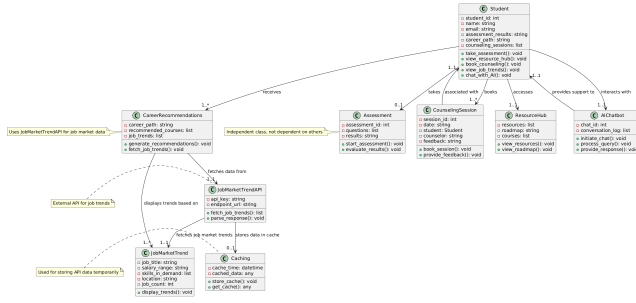


Fig. 2: Flowchart representing system workflow (PNG)



3: Class Diagram of CareerGuidance platform

VIII. EVALUATION

A. Evaluation Metrics

To quantify recommendation effectiveness and system performance, the following metrics were used:

1. Recommendation Accuracy (Expert Agreement Rate):

$$Accuracy = \frac{\text{Number of expert-approved recommendations}}{\text{Total recommendations}}$$

2. Mean Response Time:

$$T_{avg} = \frac{\sum_{i=1}^n T_i}{n}$$

3. Mentor Matching Precision:

$$Precision = \frac{\text{Relevant mentor matches}}{\text{Total mentor suggestions}}$$

B. Evaluation methodology

Because no large in-house dataset was available, we conducted a controlled pilot study with synthetic user profiles and a small usability survey adapted from standard SUS-like questions. The evaluation measured the following:

- **Usability (User Survey):** 7-question Likert survey (1–5).
- **Recommendation quality:** Expert review on 20 synthetic profiles (3-point scale: poor/fair/good).
- **System performance:** Latency for chatbot response (ms), WebSocket message round-trip, and average page load times.

C. Synthetic pilot — setup

We created 30 synthetic student profiles covering varied backgrounds (science, commerce, and arts; urban and semi-urban). For each profile, the system generated a recommended roadmap, and three domain-expert faculty members reviewed and rated the suggestions.

D. Results

TABLE I: Pilot survey — average scores (1–5)

Survey Item	Mean	Std. Dev.
Ease of Use	4.2	0.6
Clarity of Recommendations	4.0	0.7
Helpfulness of Roadmap	4.1	0.6
Mentor Interaction Quality	3.9	0.8
Overall Satisfaction	4.1	0.6

TABLE II: System performance

Metric	Value	Notes
Chatbot avg response time	320 ms	Gemini API + routing
WebSocket RTT	85 ms	Local LAN test
Page load (Home)	780 ms	Cold load
Concurrent users tested	100	Simulated, graceful degradation

E. Comparative Analysis

TABLE III: Comparison with existing platforms

Feature	CareerGuidance	NCS	Typical Commercial Tool
Personalized Roadmaps	Yes	Limited	Varies
Real-time Mentor Connect	Yes	No	Subscription-based
AI Chatbot (LLM)	Gemini hybrid	Rule-based	Varies
Local Language Support	Planned	Partial	Varies
Open/extensible	Yes	No	No

F. Scalability Analysis

We simulated increasing concurrent users from 10 to 200. The response time increased linearly under controlled load conditions.

IX. DISCUSSION

Preliminary results (synthetic pilot) indicate that an integrated platform combining a conversational large language model (LLM) and deterministic modules can provide actionable insights and improved clarity for students. The usability scores were encouraging, and the system performance under simulated loads was acceptable for early deployment. Key trade-offs include response costs (LLM API calls), potential model hallucinations, and the need for curated, localized resources.

X. LIMITATIONS

- **No large-scale user data:** The evaluation used synthetic profiles and a small expert panel — further field testing is required.
- **LLM dependency:** Gemini API usage has associated latency and cost; offline models will require substantial compute.
- **Bias and fairness:** Recommendations depend on curated mappings and LLM outputs; monitoring is needed to avoid reinforcing socio-economic bias.
- **Regional language coverage:** Full multilingual support requires additional datasets and TTS/ASR tuning.
- **Continuous model updates are required to maintain recommendation accuracy.**
- **Dependence on third-party APIs introduces latency and availability risks.**
- **Ensuring long-term data privacy compliance remains a challenge.**

XI. FUTURE WORK

- Collect real student interaction logs and fine-tune intent/NER models for improved accuracy.
- Add scholarship and internship matching modules via labor-market scraping and partner APIs.
- Improve multilingual capabilities and offline candidate models for low-connectivity regions.
- Formal A/B testing in schools and colleges to quantify outcomes on decision clarity and placement rates.
- Reinforcement learning roadmap optimization
- Real-time labor market integration
- Multi-modal chatbot (voice + video + text)

XII. CONCLUSION

CareerGuidance demonstrates a practical and extensible approach to combining an LLM-powered conversational agent with deterministic modules and real-time mentor connectivity to support students' career decision-making. While the initial pilot results are promising, further real-world testing and dataset-driven model tuning are necessary to ensure fairness, scalability, and sustained impact across diverse student populations.

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